WILDTRACK · CAPR-F2026 · GROUP 3 (TEAM 7)

Testing Report

Issue #6 · Written by Paudel Amrit (Test lead / QA) · 10 November 2026

Result: all 89 planned test cases have been run, and all 89 pass. Two cases failed on their first run, TC-66 (keyboard use, bug #13) and TC-75 (photo viewer, bug #17); both bugs were fixed and the re-tests passed. No case is blocked or pending. In two usability rounds, all 5 classmates completed all 5 tasks on their own; after the Round 1 fixes, the average score rose from 4.46 to 4.9 out of 5.

PLANNED	RUN	PASSED	BUGS FIXED	USABILITY (R2)
89	89	89	2	4.9

1. Scope and team

The tests check the functional requirements FR-01 – FR-13 and the non-functional requirements NFR-01 – NFR-08 from the test plan (`testing/TEST_PLAN.md`). The external services (GBIF, Nominatim, Photon, Wikipedia) were not tested themselves, only how WildTrack uses them and what it does when they fail.

Paudel Amrit (2530048)	Test lead. Functional tests TC-01 – TC-55 (issue #2), live-site tests TC-25, TC-26, TC-69 – TC-72, TC-77 – TC-80, this report (#6)
Gupta Aman Kumar (2530001)	Browser, phone, tablet and keyboard tests TC-58 – TC-61, TC-63 – TC-68 (issue #4); organised the usability test (#5); bug fixes (#7)
Singh Shubham Kumar (2530028)	API endpoint checks API-01 – API-09 and performance TC-56, TC-57, TC-62 (issue #3); TC-73 – TC-76, found and fixed bug #17

2. Test environment

Live site	https://wildtrack-lq88.onrender.com (Render, free plan, no database)
Local copy	XAMPP (Apache + PHP 8) on Windows, http://localhost/CAPR-F2026-WildlifeTracker/public/
Browsers	Chrome, Firefox and Edge (latest) on Windows; a phone; a narrow window (about 800 px) for tablet size
Test period	27 October – 10 November 2026 (usability Round 2 on 9 November)

3. Results by area

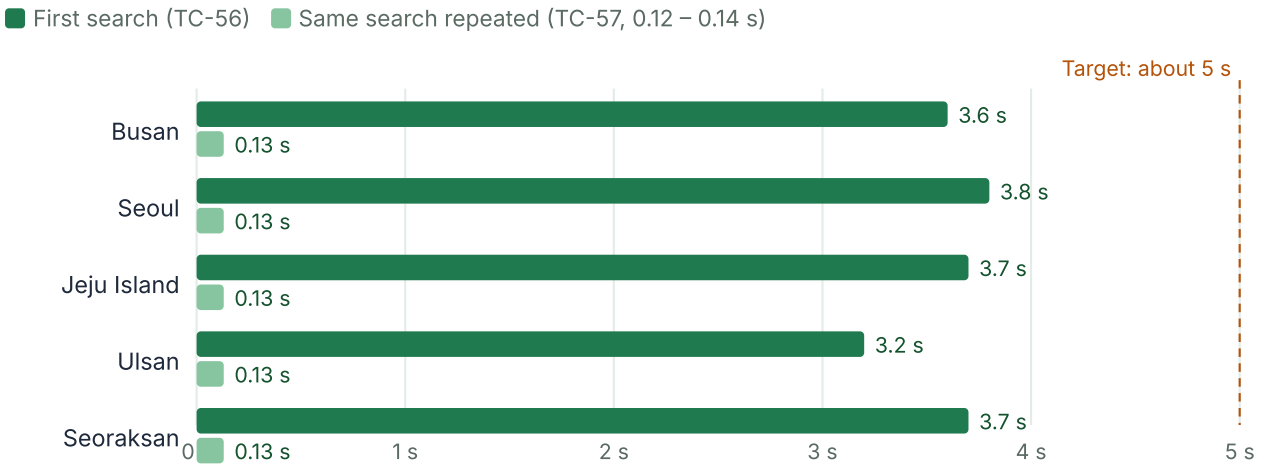
Area	Requirements	Tested by	Planned	Run	Passed
Place search, Quick Locations, Locate Me	FR-01, FR-11	Amrit	10	10	10
Map, base maps, legend	FR-02, FR-08	Amrit	4	4	4
Sightings, table, filters	FR-03 – FR-06	Amrit	14	14	14
Species Profile, GBIF link, photo viewer	FR-07, FR-09	Amrit, Shubham	8	8	8
Messages and loading	FR-10	Amrit	5	5	5

Area	Requirements	Tested by	Planned	Run	Passed
Find an Animal	FR-13	Amrit	12	12	12
Animal Encyclopedia	FR-12	Amrit, Shubham	14	14	14
Non-functional requirements	NFR-01 – NFR-08	Aman, Shubham	13	13	13
API endpoint checks	–	Shubham	9	9	9
Total			**89**	**89**	**89**

No case is marked Fail, Blocked or Pending. "Passed" includes TC-66 and TC-75, which passed after the fixes of bugs #13 and #17.

4. Performance (NFR-01)

Every first search loaded in under 4 seconds, inside the target of about 5 seconds; a repeated search loaded in about 0.13 s from the cache. Measured by Singh Shubham Kumar on the live site on 28 October (10 km, 75 records, including the place lookup).



The speed comes from two caches: the server keeps each GBIF answer for 1 hour (place names for 30 days), and the browser keeps the last 20 searches. A large first search can still take longer: Seoul at 100 km with 300 records showed the message "GBIF is slow right now — still loading..." in TC-34, as designed.

5. Bugs found and fixed

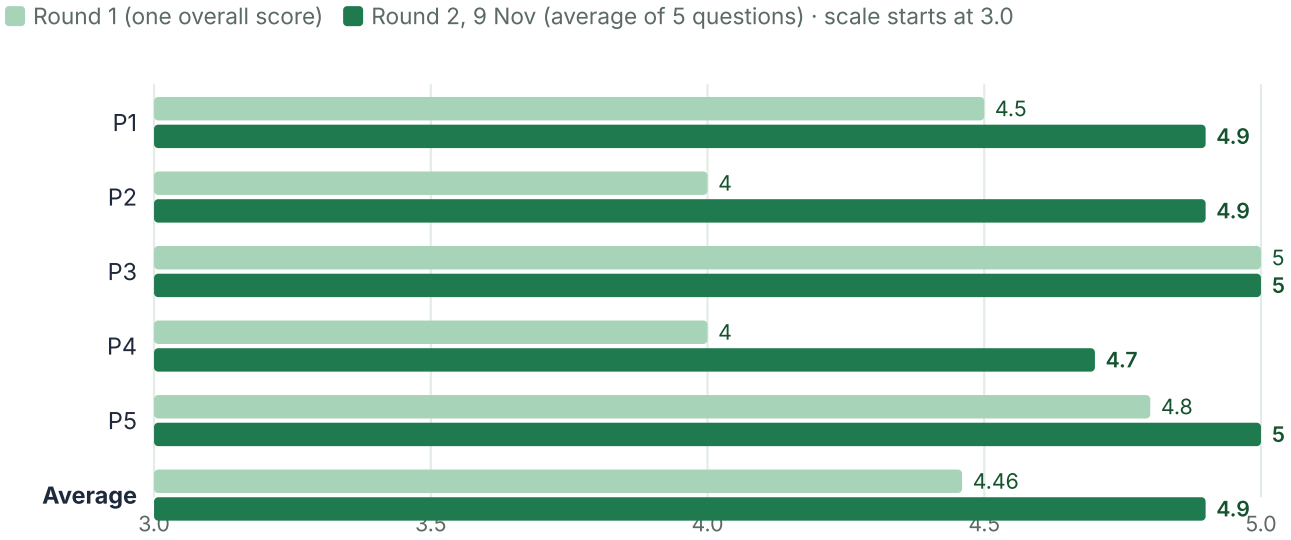
The test cases found two bugs, and both are fixed and re-tested. The functional run of TC-01 – TC-55 on 27 October found none.

Bug	Found in	Severity	Fix	Re-test
#13 – Tab to a place suggestion: no highlight, and Enter did nothing	TC-66 (keyboard)	Medium	fb5f735, 27 Oct: suggestion lists stay open when using Tab	Pass
#17 – In the photo viewer, the first photo (the sighting's own photo) had no photographer and licence; photos 2–13 had them	TC-75 (photo viewer)	Low	9c471a0, 10 Nov: the sighting photo shows its photographer and licence, e.g. "Photo: Katia (CC BY-NC-ND)"	Pass (c9d3a0a)

The usability test produced 12 further reports. They and their fixes are listed in section 6. One more fix was made during the testing phase: when a name matches several animals, Korean animals are now shown first (b931a46, 3 Nov).

6. Usability test (NFR-05)

In both rounds all 5 participants completed all 5 tasks on their own. After the Round 1 fixes, the average score rose from 4.46 to 4.9 out of 5, and every participant scored the same or higher. Five classmates from outside Group 3 (P1 – P5) took part during class breaks, organised by Gupta Aman Kumar: Round 1 on the earlier version, Round 2 on 9 November on the version with the Round 1 fixes.



Round 1: tasks, scores and suggestions

Participant	Tasks done alone	Score (of 5)	Main suggestion
P1	5 / 5	4.5	More photos in the Species Profile
P2	5 / 5	4	Search an animal without picking a species from the suggestions
P3	5 / 5	5	No suggestions after Enter or after a failed search
P4	5 / 5	4	Move the Locate Me button
P5	5 / 5	4.8	No data on the map after a failed search
Average	**25 / 25**	**4.46**	

What changed because of the usability test

The participants made 12 reports: **10 were fixed, 1 was partly fixed and 1 was not a bug.** The fixes are covered by the new test cases TC-69 – TC-77.

#	Report	Result
1	Click a photo to see it larger, with more photos	Fixed: photo viewer (TC-75, TC-76)
2	Hide suggestions after Enter; a failed search shows only "not found"	Fixed (TC-72, TC-77)
3	Choose the number of records without a dropdown	Fixed: buttons 30 / 75 / 150 / 300 (TC-25)
4	Move Locate Me to the top	Fixed (TC-26)
5	Typing "snake" or "SNA" should show all snakes	Fixed: search by a kind of animal (TC-69, TC-70, TC-73)

#	Report	Result
6	After a failed search the map should show no data	Fixed (TC-72)
7	Bigger notifications and text	Fixed
8	Order: place search, radius, Quick Locations, number of records	Fixed (TC-26)
9	Animal group ticks should follow the search ("frogs" → Amphibians)	Fixed (TC-69 – TC-71)
10	Frogs disappeared when all groups were selected	Not a bug: only the chosen number of records is loaded; the note "Showing 75 of N records" was added
11	Show whether a species may go extinct	Partly fixed: IUCN Red List status added; population trend not shown
12	Searching "reptiles" showed only one animal	Fixed (TC-71, TC-74)

Round 2: score per question (9 November)

In Round 2 every participant scored each question from 1 to 5. **Ease of use, suggestions and "would use again" scored 5.0; the lowest were knowing what the website was doing (4.7) and speed (4.8).**

Question	P1	P2	P3	P4	P5	Average
The website was easy to use.	5	5	5	5	5	**5.0**
I always knew what the website was doing (loading, results, errors).	4.5	5	5	4	5	**4.7**
The search suggestions helped me.	5	5	5	5	5	**5.0**
The results were fast enough.	5	4.5	5	4.5	5	**4.8**
I would use WildTrack again.	5	5	5	5	5	**5.0**
Average per participant	**4.9**	**4.9**	**5.0**	**4.7**	**5.0**	**4.9**

> **Limitations.** Round 1 was written down afterwards from the organiser's memory, and each person gave one overall score instead of a score per question, so Round 1 and Round 2 scores are not measured in exactly the same way. In both rounds the time per task was not measured, and the same 5 participants took part, so in Round 2 they already knew the tasks.

7. Open items

No test case is open. Two small observations remain; neither stops a main task:

Observation	Severity	Status
Some places are shown as recorded in GBIF, in Korean script ("대한민국 부산광역시 ...") or with unusual spelling ("GwanganHaebyeunr...") (TC-80)	Low	Data from GBIF, not changed
Usability Round 2: the lowest scores were "knew what the website was doing" (4.7) and speed (4.8)	Low	Possible future work

8. Conclusion

WildTrack meets its tested requirements: all 89 test cases pass, the two bugs found were fixed and re-tested, searches load within the 5-second target, and new users completed all tasks without help in both usability rounds. Round 1 led to 10 fixes, each checked again with a new test case, and the Round 2 score rose to 4.9 out of 5. The testing phase is complete.

Appendix: all test cases

Full steps, expected results and notes for every case are in `testing/TEST_PLAN.md` on GitHub (github.com/Guptaji101/CAPR-F2026-WildlifeTracker).

ID	Test	Result	Tester	Date
TC-01	Search a city	Pass	Paudel Amrit	2026-10-27
TC-02	Search button	Pass	Paudel Amrit	2026-10-27
TC-03	Empty search	Pass	Paudel Amrit	2026-10-27
TC-04	Place outside Korea	Pass	Paudel Amrit	2026-10-27
TC-05	Place suggestions	Pass	Paudel Amrit	2026-10-27
TC-06	Pick a place suggestion	Pass	Paudel Amrit	2026-10-27
TC-07	Close the suggestions	Pass	Paudel Amrit	2026-10-27
TC-08	Quick Location	Pass	Paudel Amrit	2026-10-27
TC-09	Locate Me (allowed)	Pass	Paudel Amrit	2026-10-27
TC-10	Locate Me (refused)	Pass	Paudel Amrit	2026-10-27
TC-11	Pan and zoom	Pass	Paudel Amrit	2026-10-27
TC-12	Base maps	Pass	Paudel Amrit	2026-10-27
TC-13	Recenter	Pass	Paudel Amrit	2026-10-27
TC-14	Legend	Pass	Paudel Amrit	2026-10-27
TC-15	Default search	Pass	Paudel Amrit	2026-10-27
TC-16	Table paging	Pass	Paudel Amrit	2026-10-27
TC-17	Change radius	Pass	Paudel Amrit	2026-10-27
TC-18	Apply filters	Pass	Paudel Amrit	2026-10-27
TC-19	One group	Pass	Paudel Amrit	2026-10-27
TC-20	Fish filter	Pass	Paudel Amrit	2026-10-27
TC-21	No group	Pass	Paudel Amrit	2026-10-27
TC-22	Date range	Pass	Paudel Amrit	2026-10-27
TC-23	Only a To date	Pass	Paudel Amrit	2026-10-27
TC-24	Wrong date order	Pass	Paudel Amrit	2026-10-27
TC-25	Number of records	Pass	Paudel Amrit	2026-11-04
TC-26	Number of records visible	Pass	Paudel Amrit	2026-11-04
TC-27	Select a pin	Pass	Paudel Amrit	2026-10-27
TC-28	View button	Pass	Paudel Amrit	2026-10-27
TC-29	Fish group	Pass	Paudel Amrit	2026-10-27
TC-30	GBIF link	Pass	Paudel Amrit	2026-10-27
TC-31	Close profile	Pass	Paudel Amrit	2026-10-27
TC-32	Message position	Pass	Paudel Amrit	2026-10-27
TC-33	No repeated messages	Pass	Paudel Amrit	2026-10-27
TC-34	Loading message	Pass	Paudel Amrit	2026-10-27
TC-35	Quick changes	Pass	Paudel Amrit	2026-10-27
TC-36	GBIF unreachable (local only)	Pass	Paudel Amrit	2026-10-27
TC-37	Suggestions	Pass	Paudel Amrit	2026-10-27

ID	Test	Result	Tester	Date
TC-38	Pick a suggestion	Pass	Paudel Amrit	2026-10-27
TC-39	Common name	Pass	Paudel Amrit	2026-10-27
TC-40	Scientific name	Pass	Paudel Amrit	2026-10-27
TC-41	Not recorded nearby	Pass	Paudel Amrit	2026-10-27
TC-42	Unknown name	Pass	Paudel Amrit	2026-10-27
TC-43	Show all animals	Pass	Paudel Amrit	2026-10-27
TC-44	Open page	Pass	Paudel Amrit	2026-10-27
TC-45	Every group	Pass	Paudel Amrit	2026-10-27
TC-46	Fish list	Pass	Paudel Amrit	2026-10-27
TC-47	Detail card	Pass	Paudel Amrit	2026-10-27
TC-48	Close dialog	Pass	Paudel Amrit	2026-10-27
TC-49	Wikipedia / GBIF links	Pass	Paudel Amrit	2026-10-27
TC-50	See on the map	Pass	Paudel Amrit	2026-10-27
TC-51	Search all animals	Pass	Paudel Amrit	2026-10-27
TC-52	Search all: other name	Pass	Paudel Amrit	2026-10-27
TC-53	Search in a group	Pass	Paudel Amrit	2026-10-27
TC-54	Wrong group	Pass	Paudel Amrit	2026-10-27
TC-55	Map → Encyclopedia	Pass	Paudel Amrit	2026-10-27
TC-56	NFR-01 Performance	Pass	Singh Shubham Kumar	2026-10-28
TC-57	NFR-01 Repeat	Pass	Singh Shubham Kumar	2026-10-28
TC-58	NFR-02 Chrome	Pass	Gupta Aman Kumar	2026-10-27
TC-59	NFR-02 Firefox	Pass	Gupta Aman Kumar	2026-10-27
TC-60	NFR-02 Edge	Pass	Gupta Aman Kumar	2026-10-27
TC-61	NFR-03 Security	Pass	Gupta Aman Kumar	2026-10-27
TC-62	NFR-04 Retry	Pass	Singh Shubham Kumar	2026-10-28
TC-63	NFR-05 No account	Pass	Gupta Aman Kumar	2026-10-27
TC-64	NFR-06 Phone	Pass	Gupta Aman Kumar	2026-10-27
TC-65	NFR-06 Tablet	Pass	Gupta Aman Kumar	2026-10-27
TC-66	NFR-06 Keyboard	Pass after fix	Gupta Aman Kumar	2026-10-27
TC-67	NFR-07 Free hosting	Pass	Gupta Aman Kumar	2026-10-28
TC-68	NFR-08 Structure	Pass	Gupta Aman Kumar	2026-10-28
TC-69	A kind of animal	Pass	Paudel Amrit	2026-11-04
TC-70	Snakes, also half typed	Pass	Paudel Amrit	2026-11-04
TC-71	A whole group	Pass	Paudel Amrit	2026-11-04
TC-72	Failed search empties the map	Pass	Paudel Amrit	2026-11-04
TC-73	Encyclopedia: a kind of animal	Pass	Singh Shubham Kumar	2026-11-10
TC-74	Encyclopedia: a whole group	Pass	Singh Shubham Kumar	2026-11-10
TC-75	Photo viewer	Pass after fix	Singh Shubham Kumar	2026-11-10
TC-76	Photo viewer in the Encyclopedia	Pass	Singh Shubham Kumar	2026-11-10
TC-77	A name of several animals	Pass	Paudel Amrit	2026-11-04

ID	Test	Result	Tester	Date
TC-78	Newest sightings first	Pass	Paudel Amrit	2026-11-10
TC-79	Number of records slider	Pass	Paudel Amrit	2026-11-10
TC-80	Place and record date in the profile	Pass	Paudel Amrit	2026-11-10
API-01	geocode.php?q=Ulsan	Pass	Singh Shubham Kumar	2026-10-28
API-02	geocode.php?q=Vancouver	Pass	Singh Shubham Kumar	2026-10-28
API-03	places.php?q=Jiri	Pass	Singh Shubham Kumar	2026-10-28
API-04	sightings.php? lat=35.1796&lng=129.0756&radius=10&limit=75	Pass	Singh Shubham Kumar	2026-10-28
API-05	sightings.php?...&from=2025-06-01&to=2025-01-01	Pass	Singh Shubham Kumar	2026-10-28
API-06	species.php?speciesKey=2440931	Pass	Singh Shubham Kumar	2026-10-28
API-07	animal.php?q=tiger	Pass	Singh Shubham Kumar	2026-10-28
API-08	encyclopedia.php?action=top&taxa=359&country=KR	Pass	Singh Shubham Kumar	2026-10-28
API-09	TC-61 re-check (security)	Pass	Singh Shubham Kumar	2026-10-28